CAMTECH UNIVERSITY

CAMBODIA UNIVERSITY OF TECHNOLOGY AND SCIENCE

Towards an Innovation-focused Entrepreneurial University



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ANI-TWIN

DS/AI HUB | Agriculture and Livestock Sector

A 3D Digital twin engine for virtual farm animals

Team Name: Ani-Twin

Team Members :

San Meyching	Data Science & AI	Year 1
Prithy Monynich	Data Science & AI	Year 1
Pich Soksovann	Data Science & AI	Year 1
Bin Pheakvireakbot	Software Engineering	Year 1
Toch Vanlyvey	Data Science & AI	Year 1
Po Srunheng	Data Science & AI	Year 1



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Why we started?







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Problem statement

REAL FARM

↓

NO SAFE TESTING

↓

TRAIL & ERROR

↓

COST / ANIMAL / ENVIRONMENT



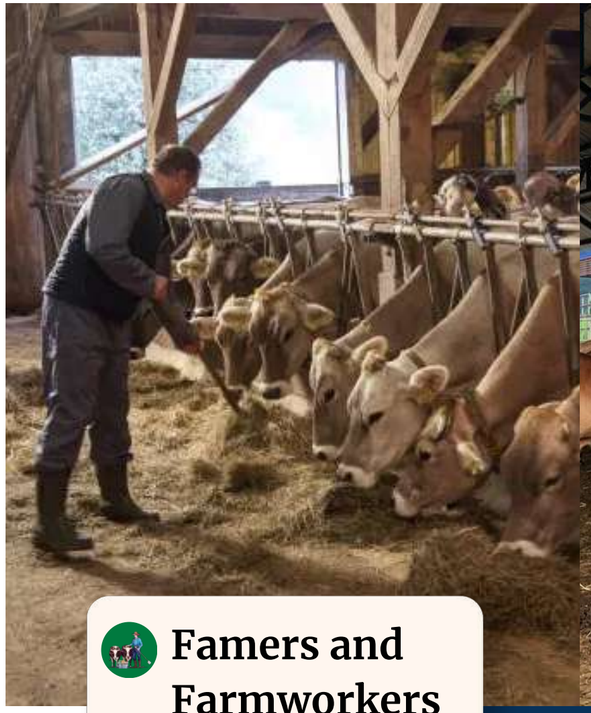
Farmers must often make changes to feed and environmental conditions on a live herd and learn from the outcome afterward. One wrong decision can mean animals stress, wasted feed and resources, lower productivity, and unnecessary environmental impact.

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Who is affected?

Primary Stakeholders



Famers and Farmworkers



Farm animals

Secondary Stakeholders



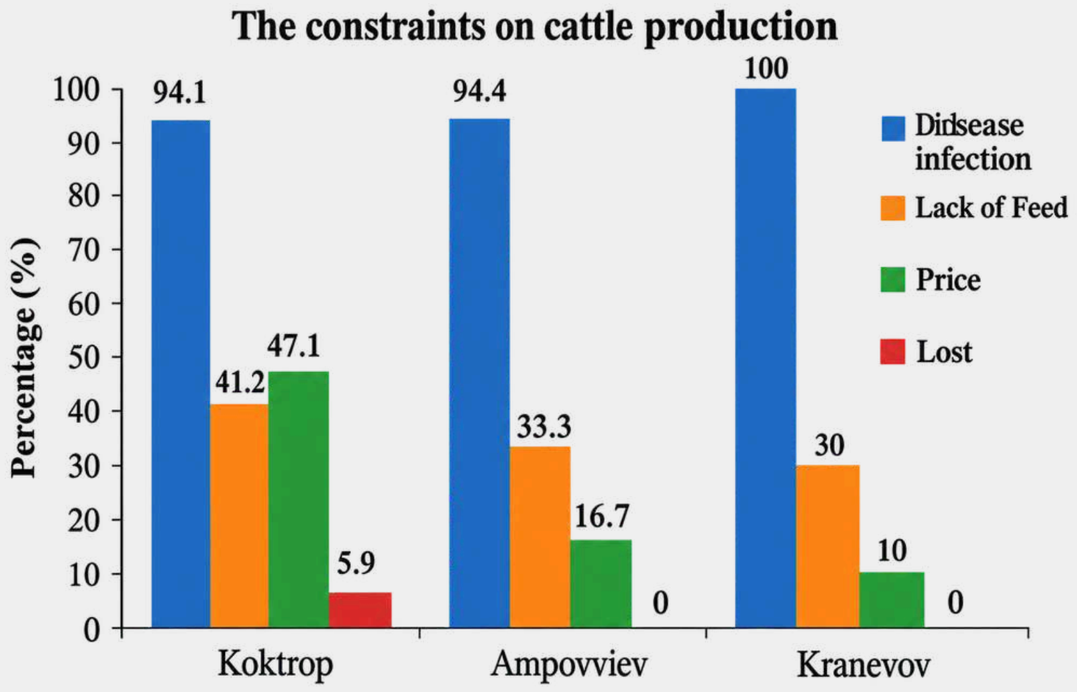
Meat Processors / Supply Chain Buyers



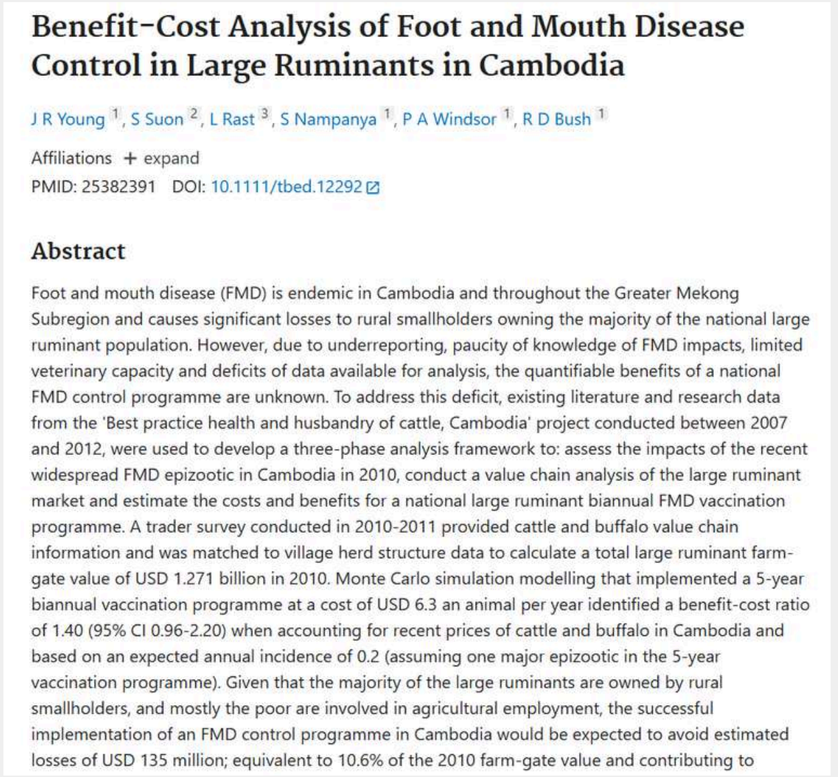
Agricultural reseachers

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Evidence



RESEARCH PAPER : Cambodia. 2012 Institute of Environmental Rehabilitation and Conservation Research Center. (2012)



Benefit–Cost Analysis of Foot and Mouth Disease Control in Large Ruminants in Cambodia

J R Young [1], S Suon [2], L Rast [3], S Nampanya [1], P A Windsor [1], R D Bush [1]

Affiliations + expand

PMID: 25382391 DOI: 10.1111/tbed.12292

Abstract

Foot and mouth disease (FMD) is endemic in Cambodia and throughout the Greater Mekong Subregion and causes significant losses to rural smallholders owning the majority of the national large ruminant population. However, due to underreporting, paucity of knowledge of FMD impacts, limited veterinary capacity and deficits of data available for analysis, the quantifiable benefits of a national FMD control programme are unknown. To address this deficit, existing literature and research data from the 'Best practice health and husbandry of cattle, Cambodia' project conducted between 2007 and 2012, were used to develop a three-phase analysis framework to: assess the impacts of the recent widespread FMD epizootic in Cambodia in 2010, conduct a value chain analysis of the large ruminant market and estimate the costs and benefits for a national large ruminant biannual FMD vaccination programme. A trader survey conducted in 2010-2011 provided cattle and buffalo value chain information and was matched to village herd structure data to calculate a total large ruminant farm-gate value of USD 1.271 billion in 2010. Monte Carlo simulation modelling that implemented a 5-year biannual vaccination programme at a cost of USD 6.3 an animal per year identified a benefit-cost ratio of 1.40 (95% CI 0.96-2.20) when accounting for recent prices of cattle and buffalo in Cambodia and based on an expected annual incidence of 0.2 (assuming one major epizootic in the 5-year vaccination programme). Given that the majority of the large ruminants are owned by rural smallholders, and mostly the poor are involved in agricultural employment, the successful implementation of an FMD control programme in Cambodia would be expected to avoid estimated losses of USD 135 million; equivalent to 10.6% of the 2010 farm-gate value and contributing to

Journal article : Benefit-Cost Analysis of Foot and Mouth Disease Control in Large Ruminants in Cambodia (2016)

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Existing Solution

Ani-Twin



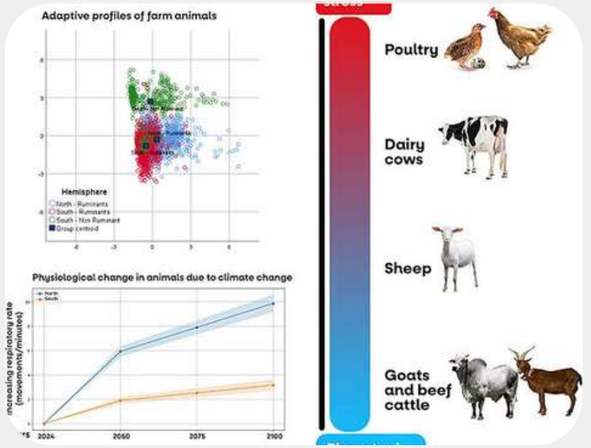
IoT based monitoring



camera/computer vison



Environmental monitoring



AI machine learning decision support system

Gap and limitation :

Does not provide a continuously updated digital representation of the animal that can be used to simulate “what-if” scenarios.

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OUR INITIAL DIRECTION

TECH STACK & BUILD APPROACH

React

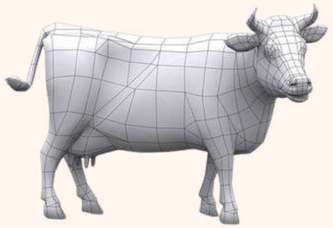
tailwindcss

Dashboard: Input data (air, temperature, diet)

PyHealth

Python: Calculate THI heat, stress

GODOT
Game engine



3D visual: make a 3D cow movement

Firebase

Vercel

Database & Punlish

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Project Timelines

Year 1 Term 3

Research & Open-Source

PROJECT PROPOSAL APPROVAL ✅ (WHERE WE ARE)



Year 2

- Term 1: 3D cow movement animation
- Term 2: Heat stress, energy balance & environmental simulation
- Term 3: Build Full system (React, Python, Firebase, Vercel)



Year 3

- Full System Performance Testing
- Validate Full System
- Web deployment
- Project Report & Final Defense

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FEASIBILITY CHECK (Risk)

Ethical

- Not Medical Advice
- Data Privacy

Main constraint

- Poor rural connectivity
- Web 3D Performance
- Hardware to train the AI model
- Formula & AI Accuracy

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Team Readiness

SUPERVISOR: VONEAT PEN
CO-SUPERVISOR: LIPHENG PRUM



Pich Soksovann: Co-Lead
Computer Vision & ML Lead



San Meyching: Team Lead
Project Lead & System Architect



Prithy Monynich
QA & System Calibration Lead



Bin Pheakvireakbot
Full-Stack Web Developer



Toch Vanlyvey
Graphics & Simulation Lead



Po SrunHeng
Biological Modeling Engineer

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Ours Research Goal



- How can digital twin technology be integrated into Cambodia livestock sector?
- What is the effect of implementing digital twins on the livestock industry?
- What are the specific situations in which digital twin technology can be utilized in the livestock industry to achieve significant advantages?

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Thank You



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“ Anitwin, testing virtually, protecting reality ”

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